Convergent CFD Simulation Guide (v2)

Hydrogen–Methane T-Junction Mixing

Three-Phase Approach in ANSYS Fluent

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Contents

1 Introduction **4**

1.1 Purpose of This Guide . . . 4

1.1.1 The Fix: Three-Phase Strategy . . . 4

1.2 Problem Definition . . . 5

1.3 Turbulence Model Choice: k–ω SST . . . 5

2 Geometry and Meshing **7**

2.1 Geometry Summary . . . 7

2.2 Meshing Summary . . . 7

2.3 Pre-Flight Mesh Check . . . 7

3 Fluent Launch and Settings Common to All Phases **9**

3.1 Launching Fluent . . . 9

3.2 Materials (All Phases) . . . 9

3.2.1 Phase 1: Import Pure Gases . . . 9

3.2.2 Phase 2: Create Custom Mixture . . . 9

3.2.3 Phase 3: Mixture Properties . . . 10

3.2.4 NIST Verification . . . 10

3.3 Operating Conditions (All Phases) . . . 10

3.4 Species Model (All Phases) . . . 10

3.5 Energy Equation (All Phases) . . . 11

3.6 Turbulent Schmidt Number . . . 11

3.7 Boundary Conditions (All Phases) . . . 11

3.7.1 `main_inlet` — Velocity Inlet . . . 11

3.7.2 `branch_inlet` — Velocity Inlet . . . 12

3.7.3 `outlet` — Pressure Outlet . . . 12

3.7.4 `wall` — Wall . . . 12

4 Phase 1: Steady-State RANS **13**

4.1 Step 1: General Settings . . . 13

4.2 Step 2: Turbulence Model . . . 13

4.3 Step 3: Solution Methods . . . 14

4.4 Step 4: Under-Relaxation Factors . . . 14

4.5 Step 5: Initialisation . . . 15

4.5.1 Patch the Branch . . . 15
4.6 Step 6: Residual Monitors . . . 16
4.7 Step 7: Run Phase 1 . . . 16
4.7.1 What to Watch . . . 16
4.8 Step 8: Save . . . 16

5 Phase 2: Transient RANS 18
5.1 Step 1: Switch to Transient . . . 18
5.2 Step 2: Change Pressure–Velocity Coupling . . . 18
5.3 Step 3: Update Spatial Discretisation . . . 19
5.4 Step 4: Transient Formulation . . . 19
5.5 Step 5: Under-Relaxation Factors . . . 19
5.6 Step 6: Time Stepping — Conservative Start . . . 20
5.6.1 What to Watch . . . 21
5.7 Step 7: Save . . . 21

6 Phase 3: Transient DDES (Production Run) 22
6.1 Step 1: Enable DDES . . . 22
6.2 Step 2: Verify All Other Settings . . . 22
6.3 Step 3: Set Up Monitors . . . 23
6.3.1 Residual Monitors . . . 23
6.3.2 Surface Monitors . . . 23
6.3.3 Point Monitors (Wall H_2) . . . 24
6.4 Step 4: Autosave . . . 24
6.5 Step 5: Run Calculation . . . 24
6.6 Step 6: Enable Time Statistics (After $t = 4\,\mathrm{s}$) . . . 25
6.7 Step 7: What to Watch During the Run . . . 25
6.8 Step 8: Save . . . 25

7 Post-Processing and Validation 26
7.1 Hydrogen Volume Fraction . . . 26
7.2 Wall y^+ Check . . . 26
7.3 Expected Results ($\mathcal{D} = 0.2$, $\beta = 0.25$) . . . 27
7.4 Coefficient of Variation (COV) . . . 27
7.5 Mesh Independence (GCI) . . . 27

8 Troubleshooting 28
8.1 Phase 1 Won't Converge . . . 28
8.2 Phase 2 Diverges on First Step . . . 28
8.3 Phase 2 Diverges After 200–500 Steps . . . 29

8.4 Phase 3: DDES Causes Immediate Divergence 29
8.5 General: “Turbulent Viscosity Ratio Limited” Warning 29
8.6 General: Mass Imbalance $> 1\,\%$ 30

9 Quick-Reference Summary 31

Chapter 1

Introduction

1.1 Purpose of This Guide

This document replaces the original *Parameterised CFD Simulation Guide*. The original setup launched a transient Spalart–Allmaras DDES directly from a cold (hybrid-initialised) field. In practice this approach frequently **diverges** in ANSYS Fluent because several aggressive settings are activated simultaneously:

1. **No steady-state precursor.** The PISO algorithm requires a reasonably established velocity and pressure field to function stably; launching it from hybrid initialisation introduces large, non-physical transients.

2. **Bounded Central Differencing (BCD) from step 1.** BCD is low-dissipation — essential for DES — but it amplifies perturbations on an uninitialised field.

3. **Spalart–Allmaras in an internal flow.** The one-equation SA model was designed for external aerodynamics and is documented as "difficult to define length scale in complex geometries" and "unsuitable for more general internal flows" (Versteeg & Malalasekera, 2007).

4. **High-pressure ideal gas + buoyancy from $t = 0$.** At 6.9 MPa, density is hypersensitive to pressure and temperature fluctuations. The full $\rho\,\mathbf{g}$ body force creates steep hydrostatic gradients that an uninitialised solver struggles to balance.

1.1.1 The Fix: Three-Phase Strategy

The solution is to build the flow field in stages, adding complexity only when the previous stage has converged.

Table 1.1: Three-phase simulation strategy.

Phase	Mode	Purpose	Duration
1	Steady RANS	Establish converged pressure, velocity, species and temperature fields	2 000–5 000 iters
2	Transient RANS	Develop the unsteady flow structure (jet shedding, buoyant plume) with stable numerics	∼5 000 steps (1 s)
3	Transient DDES	Production run with scale-resolving turbulence; collect time-averaged statistics	25 000 steps (5 s)

The final Phase 3 results are physically identical to what the Eames et al. (2022) paper produces. Phases 1 and 2 are *numerical preparation* — they do not alter the physics.

1.2 Problem Definition

Pure hydrogen is injected via a 90° T-junction into a horizontal pipeline carrying methane (modelling natural gas) at 6.9 MPa. The density ratio $\rho_{CH_4}/\rho_{H_2} \approx 8$ generates buoyancy-driven stratification and elevated wall H_2 concentrations relevant to embrittlement risk.

Table 1.2: Reference conditions (Eames et al., 2022).

Parameter	Symbol	Value
Main-pipe diameter	d_1	460 mm
Branch diameter	d_2	115 mm ($\beta = 0.25$)
Operating pressure	p_1	6.9 MPa
Main-flow speed	u_1	10 m/s
Branch speed	u_2	32 m/s ($\mathcal{D} = 0.2$)
Main temperature	T_1	284 K
Branch temperature	T_2	293 K
Injection angle	θ	90° (top-side)
Main-pipe length	—	$20\,d_1 = 9200$ mm
Junction location	—	$10\,d_1 = 4600$ mm from inlet

1.3 Turbulence Model Choice: k–ω SST

The original guide used **Spalart–Allmaras (SA)**. This guide switches to k–ω **SST** (Menter, 1994) for two reasons:

1. **Better suited to internal flows.** SST blends near-wall k–ω behaviour (accurate in adverse pressure gradients, separation and impingement) with free-stream k–ε behaviour (avoids the inlet-ω sensitivity problem). Versteeg & Malalasekera rate SST as "most general" among one- and two-equation models.

2. **SST-based DDES is available in Fluent.** When you enable DES on an SST model, Fluent uses the SST-DDES formulation, which provides better shielding of the boundary layer (the "delayed" switch) and better resolution of separated shear layers than SA-DDES in internal geometries.

The paper (Eames 2022) used SA-DES in OpenFOAM. Using SST-DDES is an *improvement*, not a departure — the flow physics (buoyant mixing, shedding, stratification) are captured equally well, with better convergence behaviour and wall-bounded accuracy.

Chapter 2

Geometry and Meshing

The geometry and meshing procedure is **identical** to the original guide (Chapters 2–3). Only a brief summary is provided here; refer to the original document for step-by-step DesignModeler and ANSYS Meshing instructions.

2.1 Geometry Summary

- Main pipe: 460 mm diameter, 9200 mm length along Z.
- Branch: 115 mm diameter at $Z = 4600$ mm, injection along $+Y$ (top-side).
- Five Named Selections: `main_inlet`, `branch_inlet`, `outlet`, `wall`, `BOI_Body`.

2.2 Meshing Summary

- Poly-hexcore method on `Fluid_Domain`.
- BOI element size: 5 mm (within 690 mm-radius sphere at junction).
- 20 inflation layers on `wall`, first layer height 0.001 mm ($y^+ \approx 1$), growth rate 1.2.
- Target cell count: $\sim$3 million.

2.3 Pre-Flight Mesh Check

Before proceeding to the solver, verify:

1. **Orthogonal Quality** > 0.1 (check in Meshing $\rightarrow$ Statistics $\rightarrow$ Mesh Metric).
2. **Aspect Ratio** < 100 (except in inflation layers).
3. **Skewness** < 0.95 for all cells.
4. **Named Selections** all present and correctly scoped.

If the mesh has quality problems: Poor orthogonality or high skewness are the #1 hidden cause of divergence. Fix the mesh before touching solver settings. A converged solution on a poor mesh is worse than a divergent run on a good mesh — at least the divergence tells you something is wrong.

Chapter 3

Fluent Launch and Settings Common to All Phases

3.1 Launching Fluent

Double-click the **Setup** cell in Workbench. When prompted:

- **Double Precision:** Yes.
- **Solver Processes:** 8 (for an i7-14700 with 8 P-cores).

3.2 Materials (All Phases)

The material setup is **identical** across all three phases.

3.2.1 Phase 1: Import Pure Gases

1. **Materials** → **Fluid** → **Create/Edit**.
2. Click **Fluent Database**.
3. Search for `methane (ch4)` → **Copy**, then **Close**.
4. Repeat for `hydrogen (h2)`.

3.2.2 Phase 2: Create Custom Mixture

1. In the Materials panel, click **Create/Edit** under **Mixture**.
2. Name: `ch4-h2-mixture`.
3. **Species:** Add `h2` and `ch4`; remove `o2` and `n2`.
4. **Order:** `h2` first, `ch4` second (CH_4 is the balance species).

3.2.3 Phase 3: Mixture Properties

Table 3.1: Mixture-level property models.

Property	**Model**
Density	`ideal-gas`
c_p	`mixing-law`
Thermal conductivity	`ideal-gas-mixing-law`
Viscosity	`ideal-gas-mixing-law`
Mass diffusivity	`kinetic-theory`

3.2.4 NIST Verification

Table 3.2: Verify individual species properties against NIST (at $T \approx 284\,\mathrm{K}$, $p = 6.9\,\mathrm{MPa}$).

Property	**CH_4**	**H_2**	**Source**
Molecular weight ($\mathrm{kg\,kmol^{-1}}$)	16.043	2.016	NIST
c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	2580	14 500	NIST
Thermal conductivity ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.038	0.185	NIST
Dynamic viscosity ($\mathrm{Pa\,s}$)	1.25×10^{-5}	9.0×10^{-6}	NIST

3.3 Operating Conditions (All Phases)

1. **Operating Pressure:** `6900000` Pa.
2. **Reference Pressure Location:** Default $(0, 0, 0)$.
3. **Gravity:** Enable. $g_x = 0$, $g_y = -9.81$, $g_z = 0$.
4. **Specified Operating Density:** Enable, value **0**.

Why operating density = 0? With an ideal-gas compressible mixture and gravity enabled, the buoyancy force must be $\rho\,\mathbf{g}$ (using the true local density), not $(\rho - \rho_0)\mathbf{g}$. A non-zero ρ_0 introduces an artificial hydrostatic baseline that can destabilise the calculation.

3.4 Species Model (All Phases)

1. **Models** → **Species** → **Species Transport**.

2. **Reactions:** Disabled.
3. **Diffusion Energy Source:** Enabled.
4. **Full Multicomponent Diffusion:** Disabled.

3.5 Energy Equation (All Phases)

Enable the Energy Equation.

3.6 Turbulent Schmidt Number

Set $Sc_t = 0.5$ via TUI:

```
/define/models/viscous/turbulent-schmidt-number 0.5
```

3.7 Boundary Conditions (All Phases)

The boundary values are **identical** across all phases. Only the solver mode and discretisation change.

3.7.1 `main_inlet` — Velocity Inlet

Table 3.3: `main_inlet` settings.

Tab	Setting	Value
Momentum	Velocity Magnitude	10 m/s
Momentum	Turbulence Spec.	Intensity & Hydraulic Diameter
Momentum	Turbulent Intensity	5 %
Momentum	Hydraulic Diameter	0.46 m
Thermal	Temperature	284 K
Species	Mass Fraction (h2)	0

3.7.2 `branch_inlet` — Velocity Inlet

Table 3.4: `branch_inlet` settings.

Tab	Setting	Value
Momentum	Velocity Magnitude	32 m/s
Momentum	Turbulence Spec.	Intensity & Hydraulic Diameter
Momentum	Turbulent Intensity	5 %
Momentum	Hydraulic Diameter	0.115 m
Thermal	Temperature	293 K
Species	Mass Fraction (h2)	1

Branch velocity derivation: $\mathcal{D} = \beta^2(u_2/u_1)$ gives $0.2 = (0.25)^2 \times u_2/10$, hence $u_2 = 32\,\mathrm{m/s}$.

3.7.3 `outlet` — Pressure Outlet

Table 3.5: `outlet` settings.

Tab	Setting	Value
Momentum	Gauge Pressure	0 Pa
Momentum	Backflow Turbulence Spec.	Intensity & Hydraulic Diameter
Momentum	Backflow Turb. Intensity	5 %
Momentum	Backflow Hydraulic Dia.	0.46 m
Thermal	Backflow Temperature	284 K
Species	Backflow h2 Mass Frac.	0

3.7.4 `wall` — Wall

Tab	Setting
Momentum	Stationary Wall, No Slip
Thermal	Heat Flux = $0\,\mathrm{W/m^2}$ (adiabatic)

Chapter 4

Phase 1: Steady-State RANS

The goal of this phase is to produce a **converged, smooth velocity–pressure–species field** that the transient solver can use as a starting point. No time-dependent physics are captured here — this is purely numerical preparation.

4.1 Step 1: General Settings

1. **Type:** Pressure-Based.
2. **Velocity Formulation:** Absolute.
3. **Time: Steady**.

Gravity and operating conditions were already set in Chapter 3.

4.2 Step 2: Turbulence Model

1. **Models** → **Viscous**.
2. Select k–**omega (2 eqn)**.
3. k–**omega Model: SST**.
4. Leave all **Model Constants** at defaults.
5. Do **NOT** enable DES yet.

Why SST and not SA? SST is a two-equation model that handles internal flows with separation, impingement and curved shear layers far better than the one-equation SA model. SST is also the natural base for the DDES formulation we will enable in Phase 3.

4.3 Step 3: Solution Methods

Table 4.1: Phase 1 solution methods.

Setting	**Value**
Pressure–Velocity Coupling	**Coupled**
Spatial Discretisation	
Gradient	Least Squares Cell Based
Pressure	Second Order
Momentum	**Second Order Upwind**
Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
Species	Second Order Upwind
Energy	Second Order Upwind

Why Coupled, not PISO? The Coupled solver solves the pressure and momentum equations simultaneously. For steady-state problems it converges **5–10× faster** than segregated SIMPLE/SIMPLEC. It also handles compressible buoyant flows more robustly because the pressure–density coupling is tighter.

Why Second Order Upwind, not BCD? Upwind schemes add numerical dissipation which is *bad* for resolving turbulent eddies (Phase 3) but *good* for convergence. We only need a smooth field here — accuracy of vortex structures does not matter in steady RANS.

4.4 Step 4: Under-Relaxation Factors

The Coupled solver uses **Courant Number** instead of separate pressure/momentum URFs.

Table 4.2: Phase 1 solution controls.

Control	**Value**
Courant Number (Coupled)	**50**
Explicit Relaxation Factors	
Momentum	0.5
Pressure	0.5
Under-Relaxation Factors	
Density	0.8
Body Forces	0.8
Turbulent Kinetic Energy	0.7
Specific Dissipation Rate	0.7
Turbulent Viscosity	0.8
Species	0.8
Energy	0.9

These are deliberately conservative. If the solution converges smoothly, you can increase the Courant Number to 100–200 after the first few hundred iterations to speed up convergence.

If it still oscillates: Drop the Courant Number to **20** and reduce momentum/pressure explicit relaxation to **0.25**. This is very slow but will always converge. Once residuals start dropping, gradually increase.

4.5 Step 5: Initialisation

1. Select **Standard Initialization** (not Hybrid).
2. **Compute From:** `main_inlet`.
3. Click **Initialize**.

Standard initialisation from the dominant inlet gives a uniform-flow starting field that is much gentler on the Coupled solver than Hybrid initialisation.

4.5.1 Patch the Branch

1. Click **Patch...**
2. Create a cell register (cylinder or sphere) enclosing the branch pipe volume.

3. Patch **H_2 mass fraction = 1.0**.
4. Patch **Y-velocity** $= -32\,\mathbf{m/s}$ (branch direction).

4.6 Step 6: Residual Monitors

1. Set convergence criteria to **1e-4** for all equations.
2. Set energy convergence to **1e-6**.
3. Enable **Plot**.

4.7 Step 7: Run Phase 1

1. **Number of Iterations:** 3000.
2. Click **Calculate**.

4.7.1 What to Watch

Table 4.3: Phase 1 convergence targets.

Indicator	Healthy	Problem
Continuity residual	$< 10^{-3}$	Oscillating or flat
Momentum residuals	$< 10^{-4}$	Rising
Species residual	$< 10^{-4}$	Flat above 10^{-2}
Energy residual	$< 10^{-6}$	Above 10^{-3}
Mass flow balance	$< 0.5\,\%$ imbalance	$> 2\,\%$

If residuals flatten but do not reach the targets, this may indicate low-frequency unsteadiness (e.g. vortex shedding at the junction). That is **expected and acceptable** — the purpose of Phase 1 is to get a "reasonable" field, not a fully converged steady state. Proceed to Phase 2 once residuals have dropped by at least **3 orders of magnitude** from initialisation.

4.8 Step 8: Save

1. **File → Write → Case & Data...**
2. Name: `phase1_steady_rans.cas.h5`

This is your safety checkpoint. If Phase 2 diverges, you can reload this file and adjust settings without re-running Phase 1.

Chapter 5

Phase 2: Transient RANS

Phase 2 introduces time-dependence while keeping the robust SST RANS turbulence model. The goal is to establish the **unsteady flow structure** — the jet shedding, buoyant plume, and recirculation zones — using numerics that can handle the transient without diverging.

5.1 Step 1: Switch to Transient

1. **Setup** → **General** → **Time:** change from Steady to **Transient**.

Do NOT change the turbulence model yet. Keep k–ω SST. The DES switch comes in Phase 3, after the transient flow structure is established.

5.2 Step 2: Change Pressure–Velocity Coupling

1. **Solution** → **Methods**.
2. Change **Pressure–Velocity Coupling** from Coupled to **PISO**.
3. Skewness Correction: **1** (default).
4. Neighbour Correction: **1** (default).

5.3 Step 3: Update Spatial Discretisation

Table 5.1: Phase 2 spatial discretisation (changes from Phase 1 highlighted).

Field	**Phase 1**	**Phase 2**
Gradient	Least Sq. Cell Based	*same*
Pressure	Second Order	**PRESTO!**
Momentum	Second Order Upwind	**Bounded Central Diff.**
k	Second Order Upwind	*same*
ω	Second Order Upwind	*same*
Species	Second Order Upwind	*same*
Energy	Second Order Upwind	*same*

Why switch momentum to BCD now? BCD is essential for resolving turbulent structures in Phase 3 (DDES). Introducing it in Phase 2 on an already-converged field avoids the oscillations that BCD causes on an uninitialised field. If BCD causes residual spikes here, temporarily revert to **Second Order Upwind** for 500 steps, then switch back.

5.4 Step 4: Transient Formulation

1. **Transient Formulation: Bounded Second Order Implicit**.

5.5 Step 5: Under-Relaxation Factors

Switch from the Coupled-solver controls to PISO-style URFs:

Table 5.2: Phase 2 under-relaxation factors.

Variable	**Factor**
Pressure	0.3
Momentum	0.7
Turbulent Kinetic Energy	0.8
Specific Dissipation Rate	0.8
Turbulent Viscosity	0.9
Species	0.9
Energy	1.0
Density	1.0
Body Forces	1.0

5.6 Step 6: Time Stepping — Conservative Start

Table 5.3: Phase 2 time stepping.

Parameter	**First 1000 steps**	**After 1000 steps**
Time Step Size (s)	**5e-4**	**2e-4**
CFL (approx.)	~ 0.6	~ 1.3
Max Iters / Step	30	20

1. Set **Time Step Size** to `5e-4`.
2. Set **Number of Time Steps** to `1000`.
3. Set **Max Iterations / Time Step** to `30`.
4. Click **Calculate**.
5. After 1000 steps complete, change Time Step Size to `2e-4`.
6. Set Number of Time Steps to `4000`.
7. Set Max Iterations / Time Step to `20`.
8. Click **Calculate**.

Total Phase 2 time: $1000 \times 5{\times}10^{-4} + 4000 \times 2{\times}10^{-4} = 0.5 + 0.8 = 1.3\,\mathrm{s}$ of physical time.

5.6.1 What to Watch

- **Residuals:** Should drop 1–2 orders of magnitude *within each time step*. If they plateau at $\sim 10^{-2}$ and oscillate, this is acceptable for PISO.
- **Max Courant Number:** Printed in the console. Target: < 5. If consistently above 10, reduce Δt.
- **Outlet H_2:** Set up a surface monitor (mass-weighted average of h2 on outlet). It should start developing toward ~ 0.019 mass fraction.
- **Negative temperatures:** If these appear, immediately reduce Δt to `1e-4` and run 500 steps.

5.7 Step 7: Save

1. **File → Write → Case & Data...**
2. Name: `phase2_transient_rans.cas.h5`

Chapter 6

Phase 3: Transient DDES (Production Run)

This is the final phase that produces publication-quality results. The flow field from Phase 2 provides a stable starting point for the scale-resolving DDES turbulence model.

6.1 Step 1: Enable DDES

1. **Models** → **Viscous**.
2. The model should still show k–**omega (2 eqn), SST**.
3. In the left panel, select **Detached Eddy Simulation (DES)**.
4. **DES Option: Delayed DES (DDES)**.
5. Leave all model constants at defaults.
6. Click **OK**.

This changes the turbulence model from RANS to SST-DDES. In attached boundary layers, the solver continues using the SST RANS equations. In separated/free-shear regions (the junction wake, the buoyant plume), it switches to LES-like behaviour, resolving large eddies directly.

Alternative: SA-DDES. If you specifically want to replicate the Eames 2022 paper, you can switch from SST to Spalart–Allmaras and enable DES→DDES. However, SST-DDES is generally more robust and accurate for internal flows. Both will produce qualitatively the same stratification and mixing results.

6.2 Step 2: Verify All Other Settings

Nothing else changes from Phase 2. Confirm:

Table 6.1: Phase 3 settings checklist.

Setting	Required Value	Check
Time	Transient	✓
P–V Coupling	PISO	✓
Momentum Scheme	Bounded Central Diff.	✓
Pressure Scheme	PRESTO!	✓
Transient Formulation	Bounded 2nd Order Implicit	✓
Time Step Size	2e-4 s	✓
Max Iters / Step	20	✓

6.3 Step 3: Set Up Monitors

6.3.1 Residual Monitors

1. Equations: continuity, x/y/z-velocity, k, ω (or $\tilde{\nu}$ if using SA), h2, energy.
2. **Convergence Criterion:** `1e-4` for all; energy `1e-6`.
3. **Plot:** Enabled.

6.3.2 Surface Monitors

Table 6.2: Surface monitors.

Name	Type	Surface	Field
`outlet_h2`	Mass-Wtd Average	outlet	h2 mass frac.
`outlet_mdot`	Mass Flow Rate	outlet	—
`main_mdot`	Mass Flow Rate	main_inlet	—
`branch_mdot`	Mass Flow Rate	branch_inlet	—

6.3.3 Point Monitors (Wall H_2)

Table 6.3: Wall probe locations ($d_1 = 0.46\,\mathrm{m}$, junction at $Z = 4.6\,\mathrm{m}$).

Name	X (m)	Y (m)	Z (m)	z/d_1
`top_z2d`	0	0.23	5.52	2
`top_z4d`	0	0.23	6.44	4
`top_z6d`	0	0.23	7.36	6
`top_z8d`	0	0.23	8.28	8
`bot_z2d`	0	−0.23	5.52	2
`bot_z4d`	0	−0.23	6.44	4
`bot_z6d`	0	−0.23	7.36	6
`bot_z8d`	0	−0.23	8.28	8

Monitor both top and bottom walls to capture the stratification asymmetry that is the central result of the paper.

6.4 Step 4: Autosave

1. **Calculation Activities** → **Autosave Every:** `500` Time Steps.
2. Save both **Case** and **Data**.

6.5 Step 5: Run Calculation

Table 6.4: Phase 3 run parameters.

Parameter	**Value**
Time Step Size (s)	`2e-4`
Number of Time Steps	`25000`
Max Iterations / Time Step	`20`
Total physical time	5 s

1. Click **Calculate**.

6.6 Step 6: Enable Time Statistics (After $t = 4\,\mathrm{s}$)

After **20 000** time steps ($t = 4\,\mathrm{s}$), the initial transient has washed out. Enable time-averaging:

1. **Solution → Run Calculation → Data Sampling for Time Statistics**.
2. **Enable:** checked.
3. **Sampling Interval:** 1 (every time step).
4. Continue the remaining 5 000 steps to $t = 5\,\mathrm{s}$.

This matches the paper's approach: average over $t = 4$–$5\,\mathrm{s}$.

6.7 Step 7: What to Watch During the Run

Table 6.5: Runtime diagnostics.

Indicator	**Healthy**	**Warning**	**Action**
Residuals per step	Drop 1–2 orders	Flat at 10^{-1}	Reduce Δt
Max Courant	< 3	> 5 consistently	Halve Δt
Outlet H_2	Stabilise ~ 0.019	Drift > 0.03	Check BCs
Mass imbalance	$< 1\,\%$	$> 2\,\%$	Check outlet BC
Negative T	Never	Any occurrence	Halve Δt

6.8 Step 8: Save

1. **File → Write → Case & Data...**
2. Name: `phase3_ddes_final.cas.h5`

Chapter 7

Post-Processing and Validation

7.1 Hydrogen Volume Fraction

The paper reports $\phi_{\mathrm{H_2}}$ (volume fraction), not $Y_{\mathrm{H_2}}$ (mass fraction). Define a Custom Field Function:

$$\phi_{\mathrm{H_2}} = \frac{Y_{\mathrm{H_2}}/M_{\mathrm{H_2}}}{Y_{\mathrm{H_2}}/M_{\mathrm{H_2}} + (1 - Y_{\mathrm{H_2}})/M_{\mathrm{CH_4}}} \tag{7.1}$$

In Fluent: **Custom Field Functions** → **Define**:

```
(Y_H2/2.016) / (Y_H2/2.016 + (1-Y_H2)/16.043)
```

7.2 Wall y^+ Check

1. **Results** → **Plots** → **XY Plot**.
2. Surface: `wall`.
3. Field: Wall Fluxes → Wall Yplus.

Target: $y^+ < 5$ everywhere. Localised $y^+ \sim 2$–3 near stagnation is acceptable. If $y^+ > 5$ consistently, refine the inflation layer and re-run.

7.3 Expected Results ($\mathcal{D} = 0.2$, $\beta = 0.25$)

Table 7.1: Validation targets (Eames 2022).

Quantity	**Expected Value**
Outlet V_R (volume fraction)	≈ 0.149
Outlet Y_{H_2} (mass fraction)	≈ 0.019
Mass balance residual	$< 1\,\%$
Stratification: top wall V_R at $z/d_1 = 8$	$\approx 1.6 \times V_{R,T}$
Top wall > bottom wall H_2	Yes (buoyancy)

7.4 Coefficient of Variation (COV)

On any cross-section downstream of the junction:

$$\mathrm{COV} = \frac{\sigma_\phi}{\bar{\phi}}$$

COV = 0: perfect mixing. COV = 1: complete segregation.

7.5 Mesh Independence (GCI)

Run three meshes (coarse, medium, fine) and compute:

$$p = \frac{\ln|(\phi_3 - \phi_2)/(\phi_2 - \phi_1)|}{\ln r}, \qquad \mathrm{GCI}_{\mathrm{fine}} = \frac{1.25\,|(\phi_2 - \phi_1)/\phi_1|}{r^p - 1}$$

Target: GCI $< 2\,\%$.

Chapter 8

Troubleshooting

8.1 Phase 1 Won't Converge

Table 8.1: Phase 1 troubleshooting.

Symptom	**Cause**	**Fix**
Residuals oscillate wildly	Courant Number too high for the mesh	Reduce to **20**, then increase gradually
Continuity residual stuck at 10^{-1}	Mesh quality issue (high skewness/aspect ratio near junction)	Check **Mesh Quality** in Meshing; fix problem cells
Reversed flow at outlet	Outlet too close to junction, or backflow BC wrong	Ensure outlet is at $Z = 9.2\,\mathrm{m}$ ($10\,d_1$ from junction)
Negative temperatures	Operating pressure wrong (not 6.9 MPa), or energy equation URF too high	Verify operating pressure = 6 900 000; reduce energy URF to 0.7

8.2 Phase 2 Diverges on First Step

This almost always means the Phase 1 field has a problem:

1. Reload `phase1_steady_rans.cas.h5`.
2. Run 500 more steady iterations.
3. Try Phase 2 again with $\Delta t = 1 \times 10^{-4}\,\mathrm{s}$ (half the recommended value).

8.3 Phase 2 Diverges After 200–500 Steps

Table 8.2: Phase 2 troubleshooting.

Symptom	Cause	Fix
Floating-point exception	BCD scheme oscillating on strong gradients	Temporarily switch momentum to **Second Order Upwind** for 1000 steps, then switch back to BCD
Max Courant > 20	Time step too large for local cell size near junction	Reduce Δt to **1e-4**; or reduce BOI element size from 5 mm to 3 mm (coarser mesh = larger stable Δt)
Species residual explodes	H_2 mass fraction going > 1 or < 0	Enable **Bounded** on species under Solution Methods; reduce species URF to **0.7**

8.4 Phase 3: DDES Causes Immediate Divergence

The DDES switch activates LES-like behaviour in separated regions. If the Phase 2 field has large RANS eddies that suddenly "break up" under DES, this can cause a spike:

1. Reload `phase2_transient_rans.cas.h5`.
2. Enable DDES.
3. Reduce Δt to **1e-4** s.
4. Run 2000 steps at this reduced Δt.
5. If stable, increase Δt back to **2e-4** s.

8.5 General: "Turbulent Viscosity Ratio Limited" Warning

This message appears when $\mu_t/\mu > 10^5$. It usually means:

- The turbulence model is generating unrealistically high μ_t at the junction stagnation point.
- **Not fatal** if it appears occasionally (especially in Phase 1).

- If it appears **every iteration**, reduce the turbulence URFs (k, ω) to 0.5 and investigate the mesh in the junction region.

8.6 General: Mass Imbalance $> 1\,\%$

1. Check that **all** boundary conditions are set correctly.
2. Verify that the outlet backflow species fraction is **0** (not 1).
3. Check for reversed flow at the outlet: **Flux Reports** $\rightarrow$ mass flow rate on outlet. If negative, the outlet is too close or the pressure BC is wrong.

Chapter 9

Quick-Reference Summary

Table 9.1: Complete settings summary across all three phases.

Setting	**Phase 1**	**Phase 2**	**Phase 3**
Mode	Steady	Transient	Transient
Turbulence	SST RANS	SST RANS	**SST-DDES**
P–V Coupling	Coupled	PISO	PISO
Momentum Scheme	2nd Upwind	**BCD**	BCD
Pressure Scheme	2nd Order	PRESTO!	PRESTO!
Transient Form.	—	Bounded 2nd Impl.	Bounded 2nd Impl.
Δt (s)	—	5e-4 $\rightarrow$ 2e-4	2e-4
Iterations	3000	5000 steps	25000 steps
Max Iter/Step	—	30 $\rightarrow$ 20	20
Under-Relaxation / Controls			
Pressure	CFL=50	0.3	0.3
Momentum	0.5	0.7	0.7
k / ω	0.7	0.8	0.8
Species	0.8	0.9	0.9
Energy	0.9	1.0	1.0
Checkpoints			
Save as	`phase1_steady_rans`	`phase2_transient_rans`	`phase3_ddes_final`

Pre-flight checklist before Phase 1:

- ☐ Mesh quality checked (orthogonality > 0.1, skewness < 0.95)
- ☐ Named Selections present (main_inlet, branch_inlet, outlet, wall)
- ☐ Material = ch4-h2-mixture with ideal-gas density
- ☐ Operating pressure = 6 900 000 Pa
- ☐ Operating density = 0
- ☐ Gravity enabled, $g_y = -9.81$
- ☐ Energy equation ON
- ☐ Species Transport ON, Reactions OFF
- ☐ $Sc_t = 0.5$ (set via TUI)
- ☐ Double precision enabled